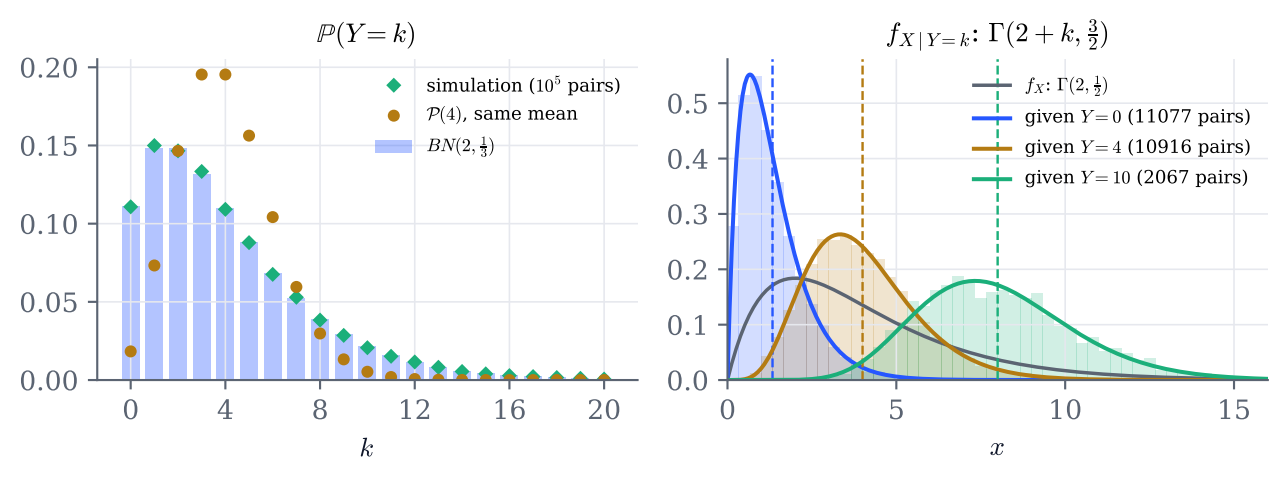
$\mathbb{P}(Y=k)$
simulation (10^5 pairs)
$\mathcal{P}(4)$, same mean
$BN(2,\frac{1}{3})$
0.20
0.15
0.10
0.05
0.00
0
4
8
12
16
20
k
$f_{X\,|\,Y=k}$: $\Gamma(2+k,\frac{3}{2})$
f_X: $\Gamma(2,\frac{1}{2})$
given $Y=0$ (11077 pairs)
given $Y=4$ (10916 pairs)
given $Y=10$ (2067 pairs)
0.5
0.4
0.3
0.2
0.1
0.0
0
5
10
15
x